eAuto – Bike Spare Parts E-Commerce Platform



1. Introduction
2. Aim of the Project
3. Features of the Project
4. System Analysis & Design
5. ER Diagram
6. Data Flow Diagram (DFD)
7. Flow Charts
8. Implementation Details
9. Testing
10. Limitations of the Project
11. Conclusion
12. References

Project Introduction

Background of the Project – Short description of the problem domain and why the project was initiated.

Purpose of the Project – What the system aims to achieve and its importance.

Scope of the Project – Boundaries of the system, what it will and will not cover.

Project Benefits

Objectives – Clear, measurable goals of the system.

Advantages of the System – Key benefits like automation, user convenience, admin control, etc.

Aim of the Project

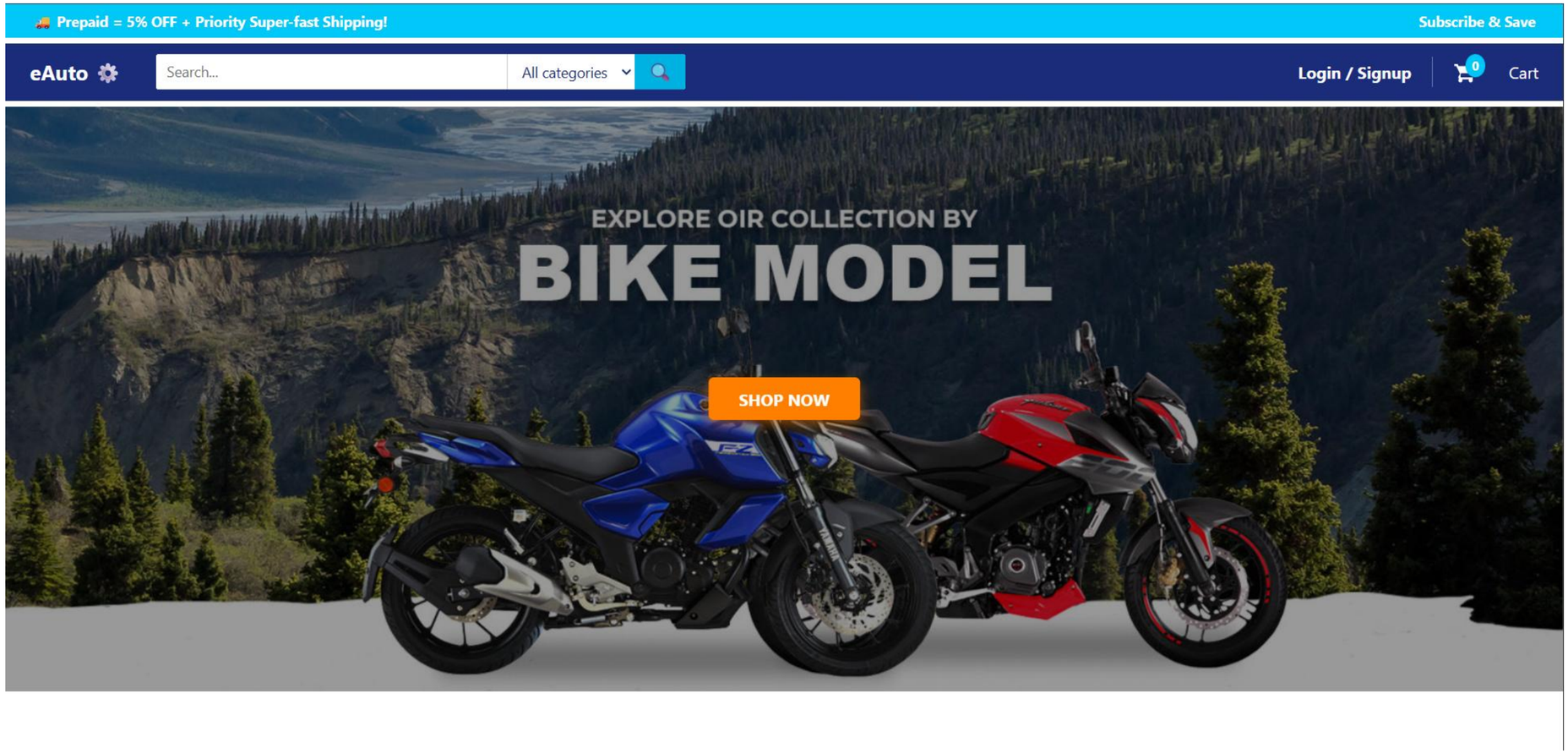
Statement of Aim – Clearly defines what the project intends to achieve.

Importance of Aim – Explains why the aim is necessary and valuable.

How Aim Solves Current Problems – Shows how the project overcomes existing system drawbacks.

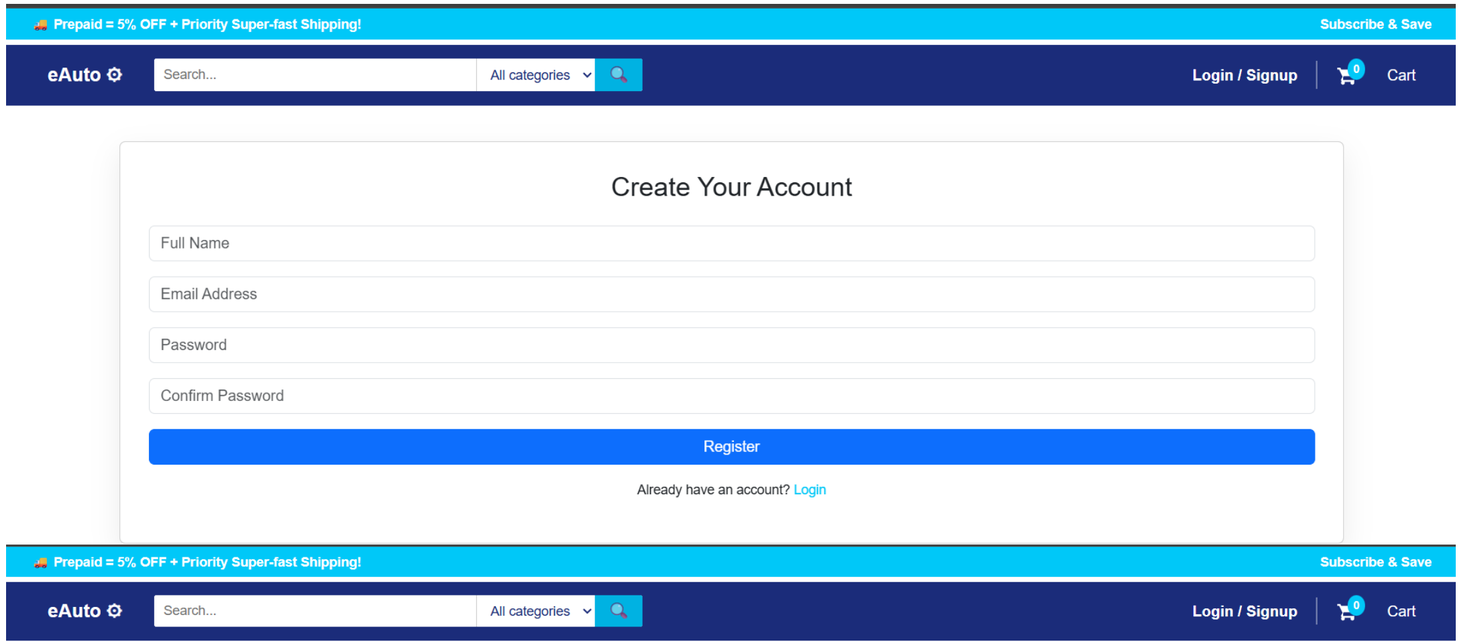
Features of the Project

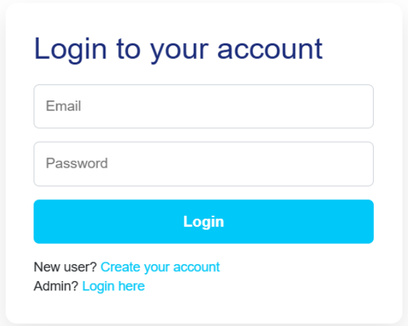
Content: Brief overview → “The system provides features for both Users and Admins to ensure smooth e-commerce operations.”



User Feature1: Registration & Login

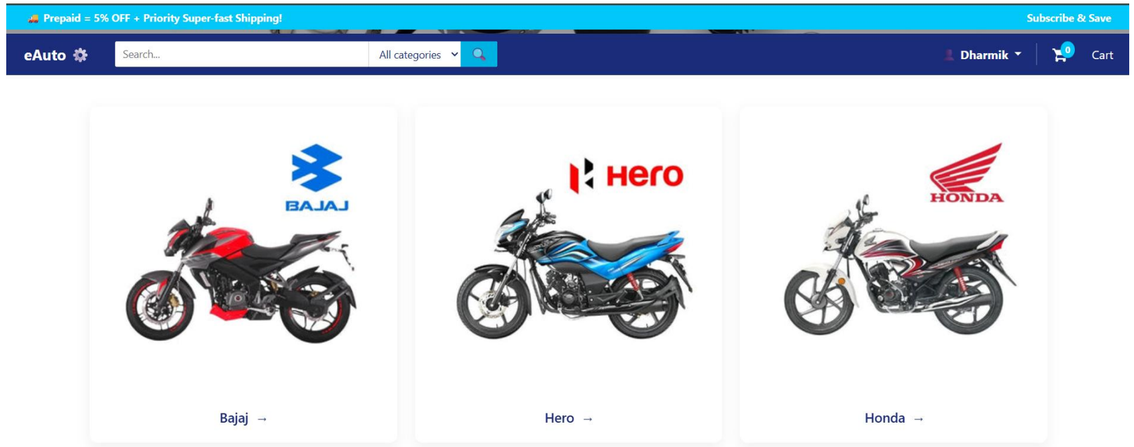
- Secure authentication
- User session management

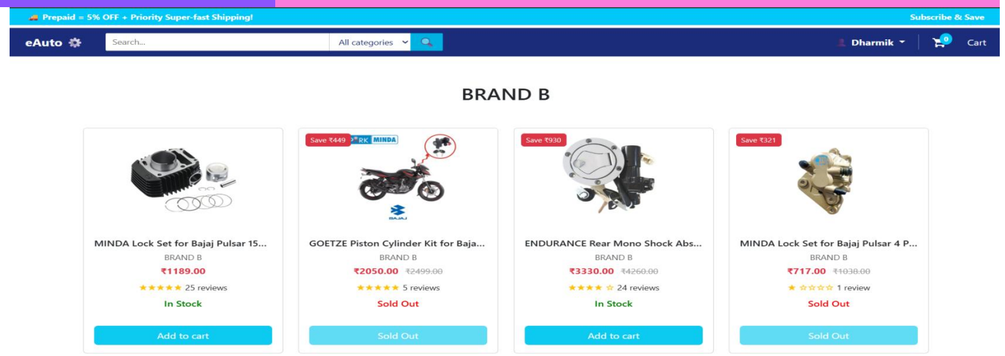


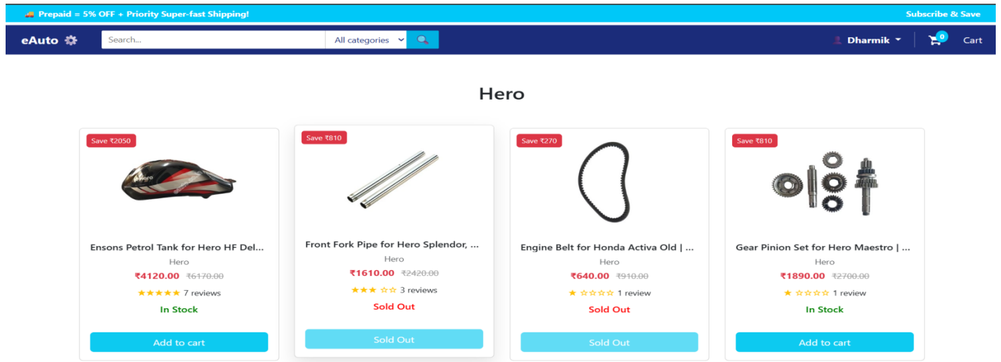


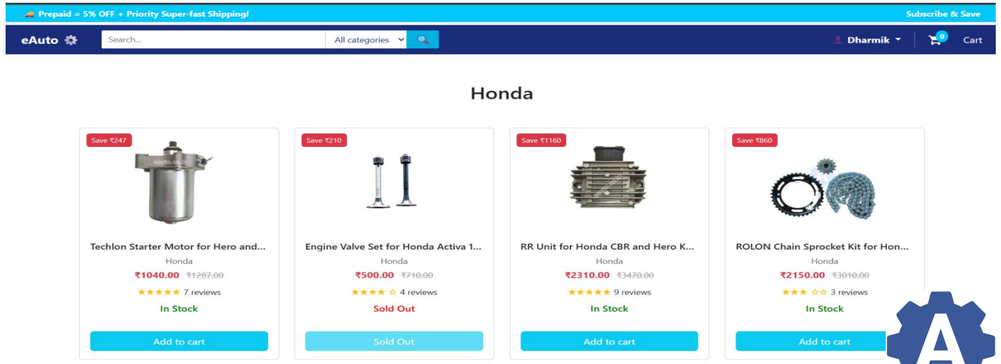
User Feature 2: Browse & Search Products

- Search by brand/category
- Product filtering & sorting



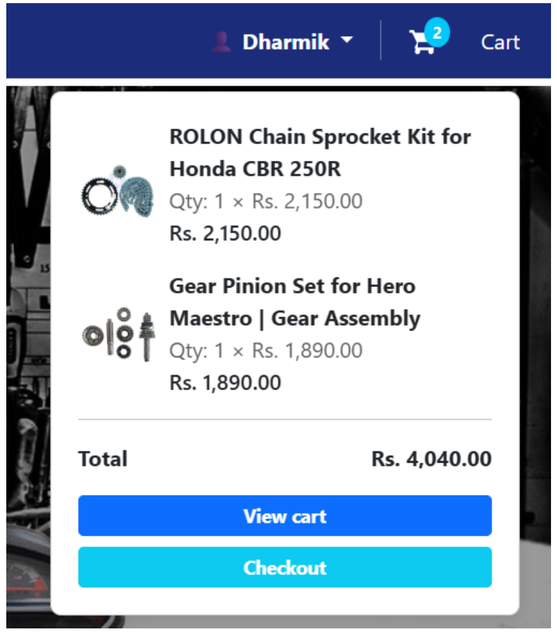


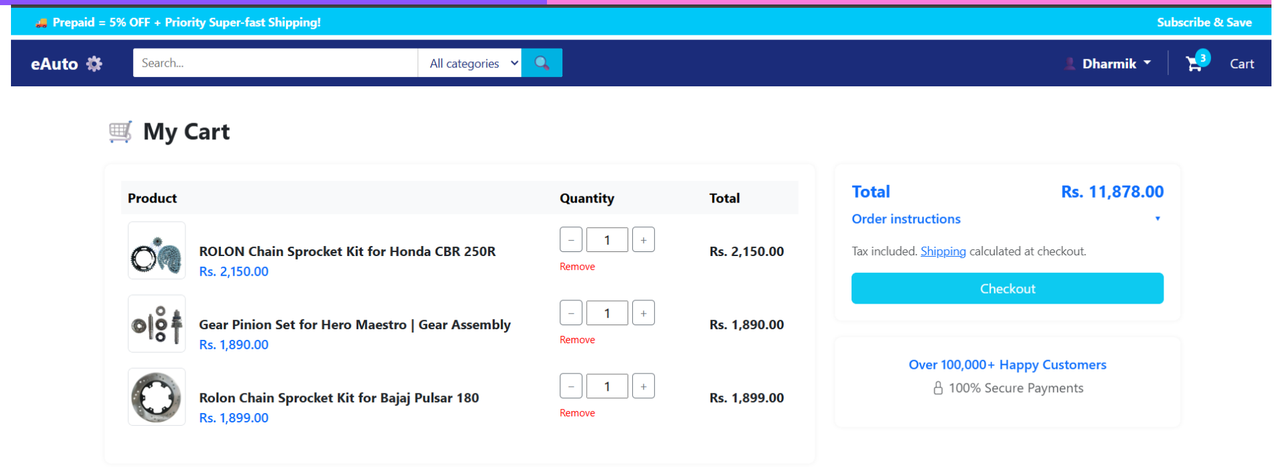


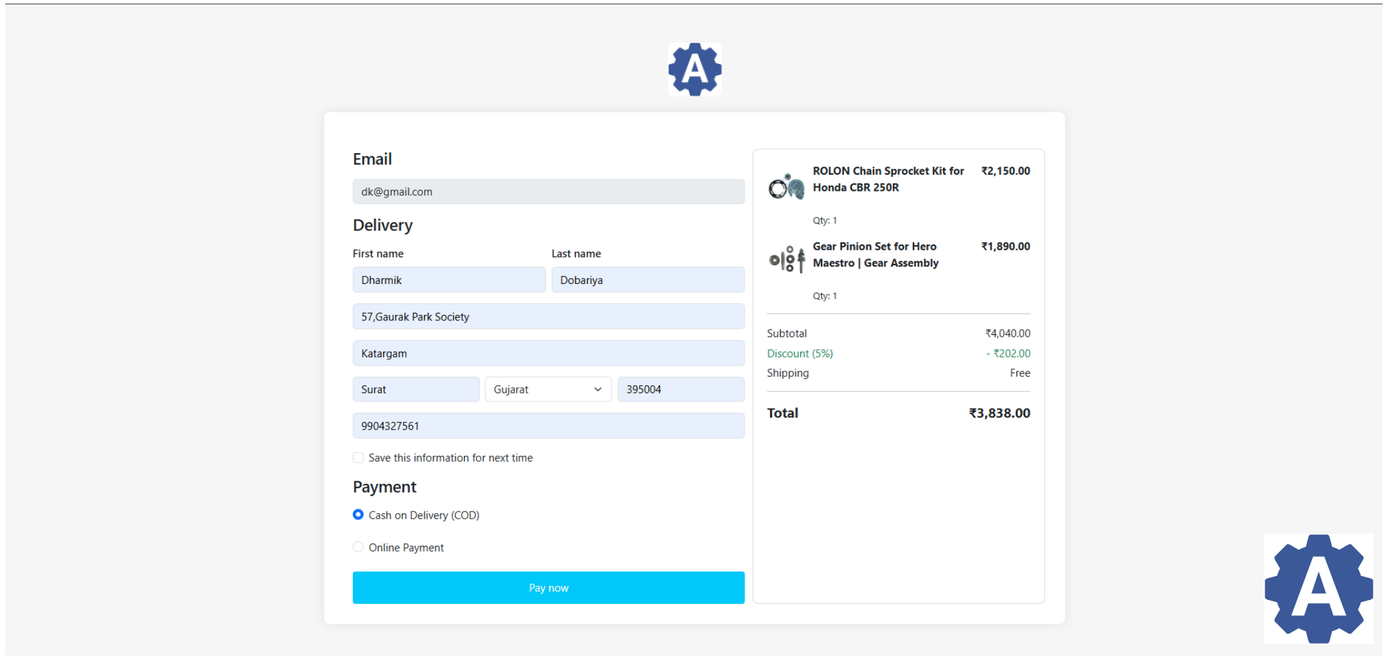


User Feature 3: Cart & Checkout System

- Add/remove products from cart
- Secure checkout process

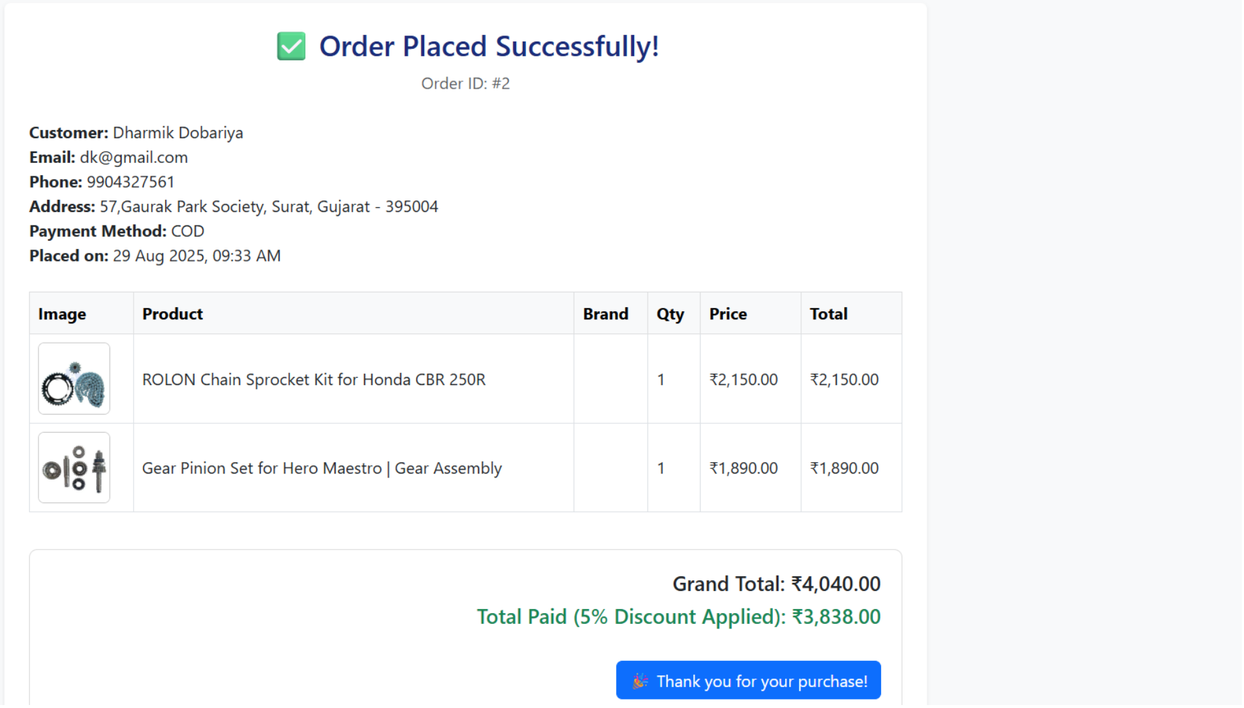






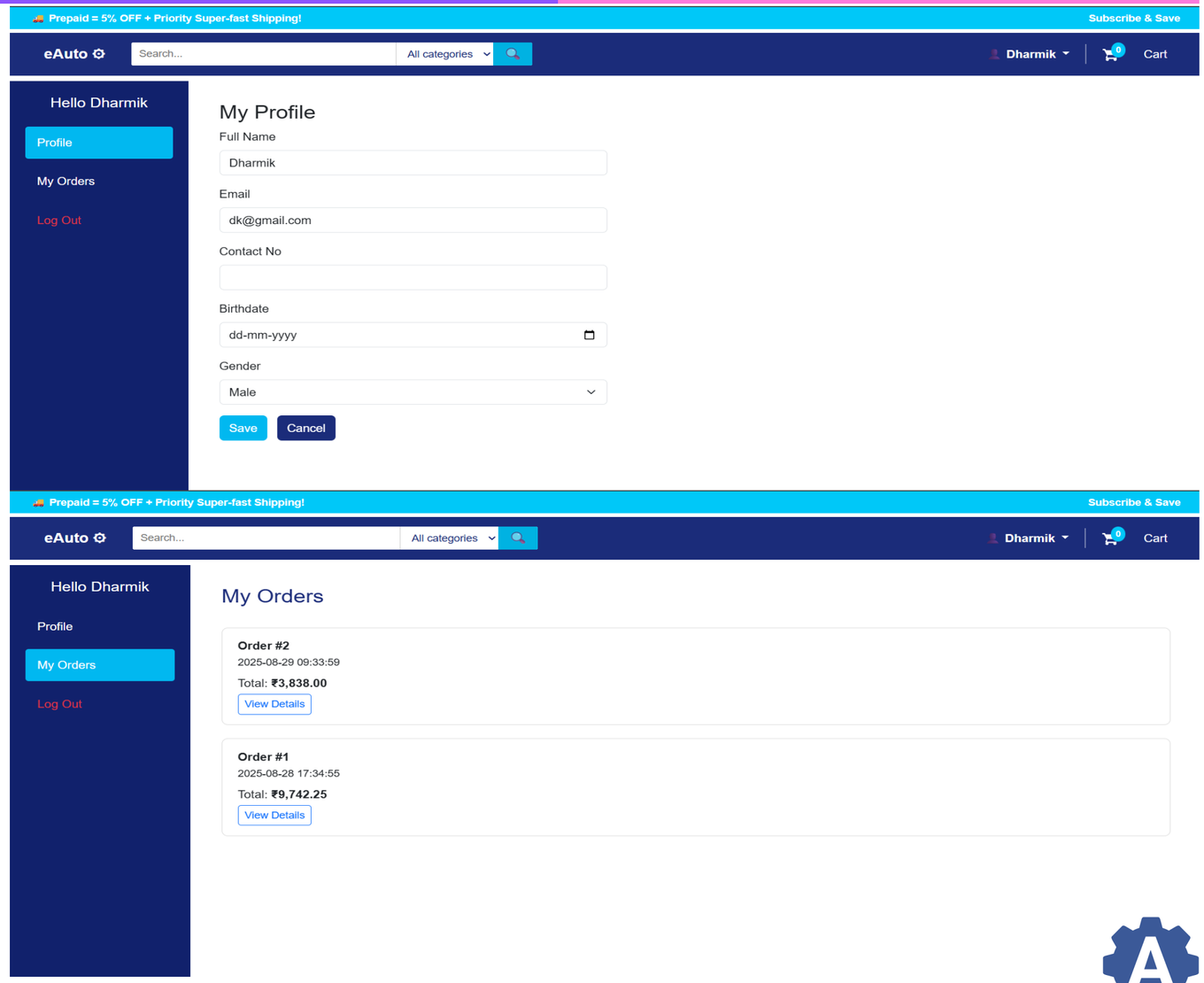
User Feature 4: Order Tracking

- Real-time order status
- Delivery updates



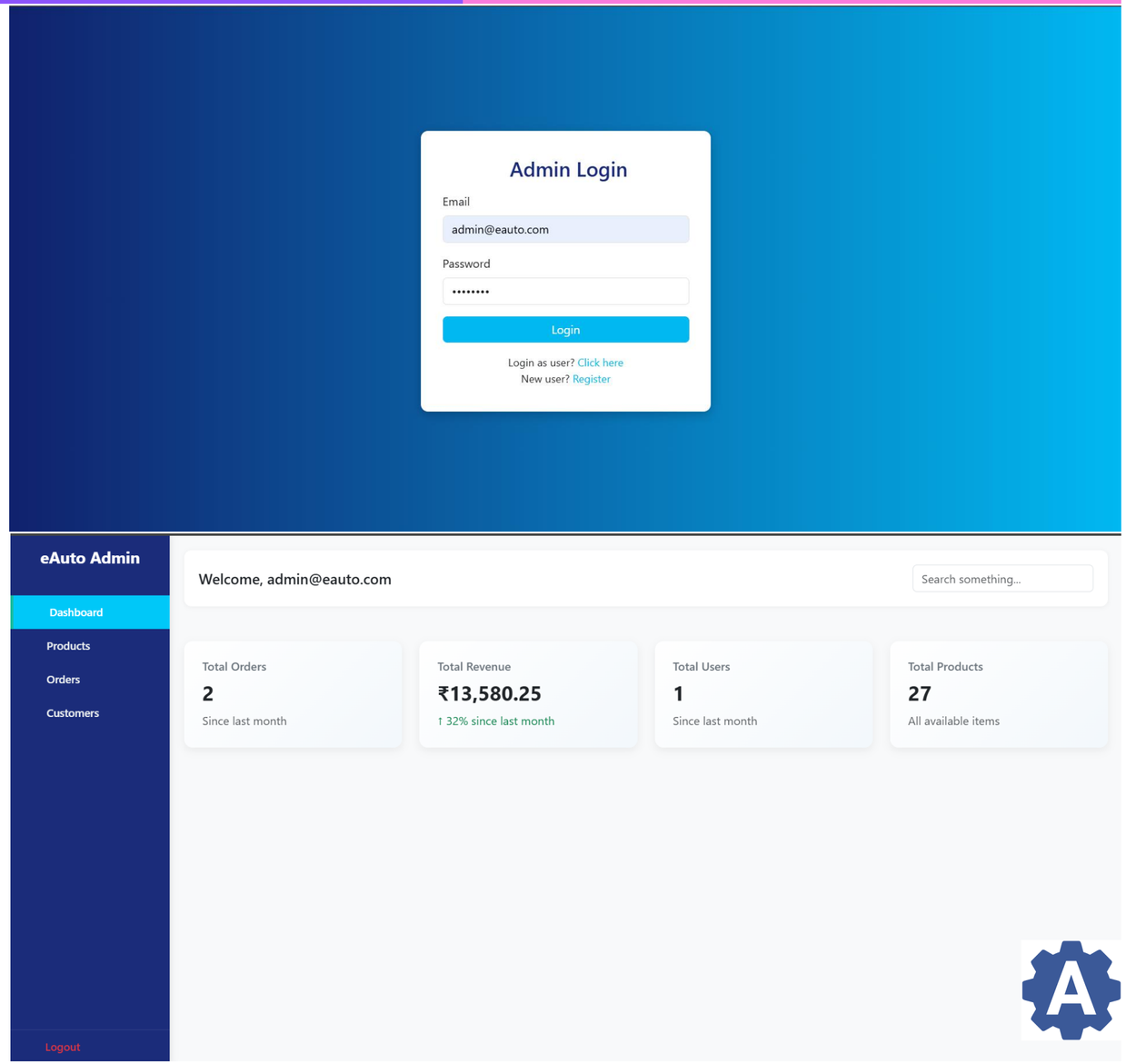
User Feature 5: Profile & My Orders

- Manage personal info
- View details my orders



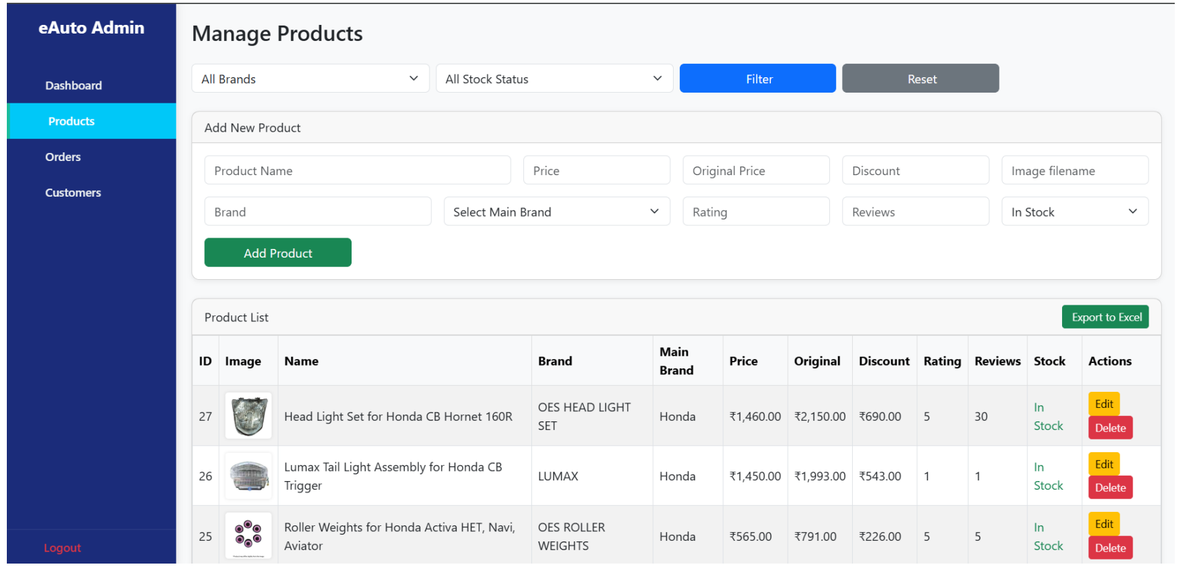
Admin Feature 1: Admin Login & Dashboard

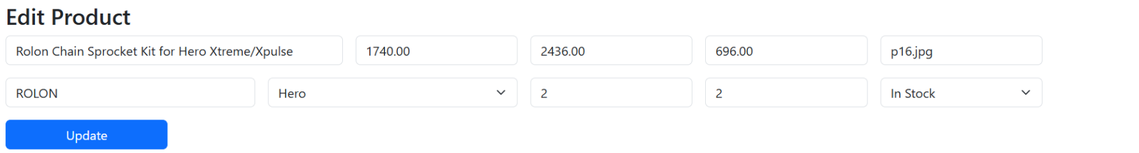
- Secure admin authentication
- Centralized dashboard view

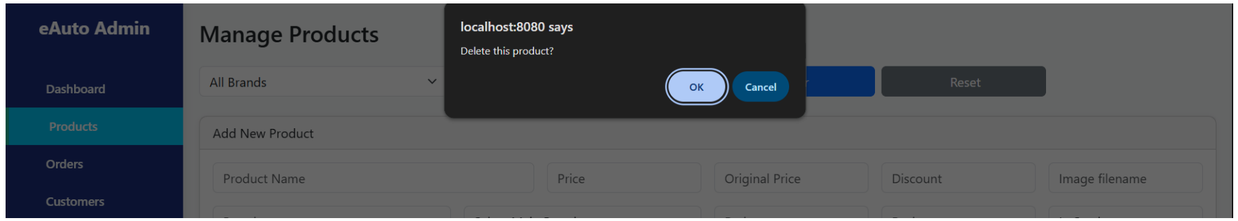


Admin Feature 2: Manage Products (Add/Edit/Delete)

- CRUD operations on products
- Product stock & pricing control

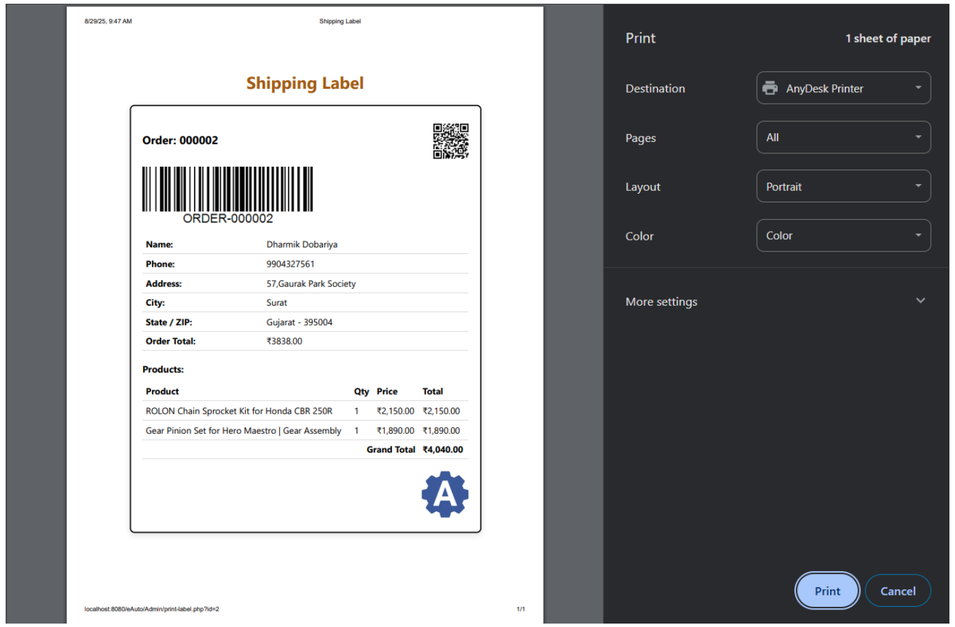


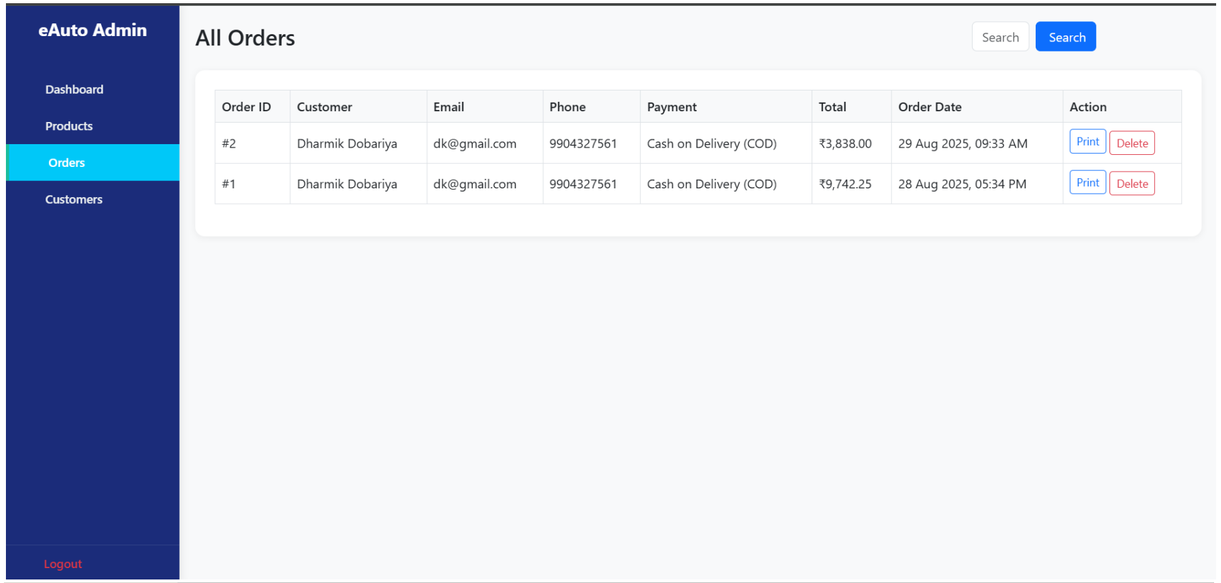


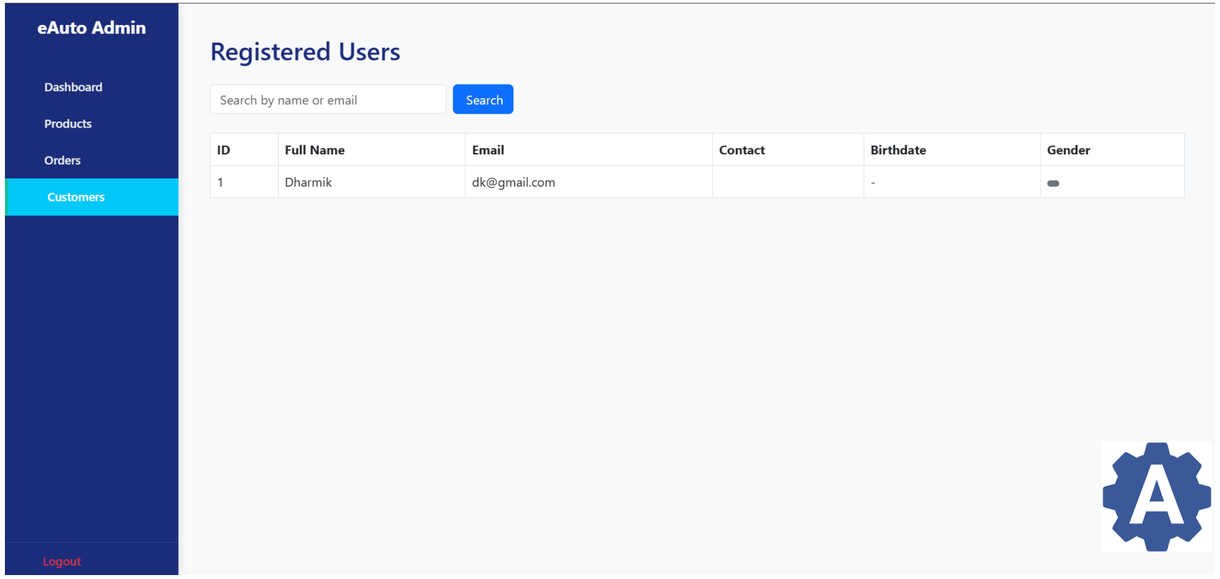


Admin Feature 3: Manage Orders & Users

- View/approve/reject orders
- Manage user details & roles







System Analysis

- **Requirement Gathering** – Collecting needs from users & admin.
- **Functional Requirements** – Features the system must provide (e.g., login, cart, product management).
- **Non-Functional Requirements** – Performance, security, scalability, usability.

System Design

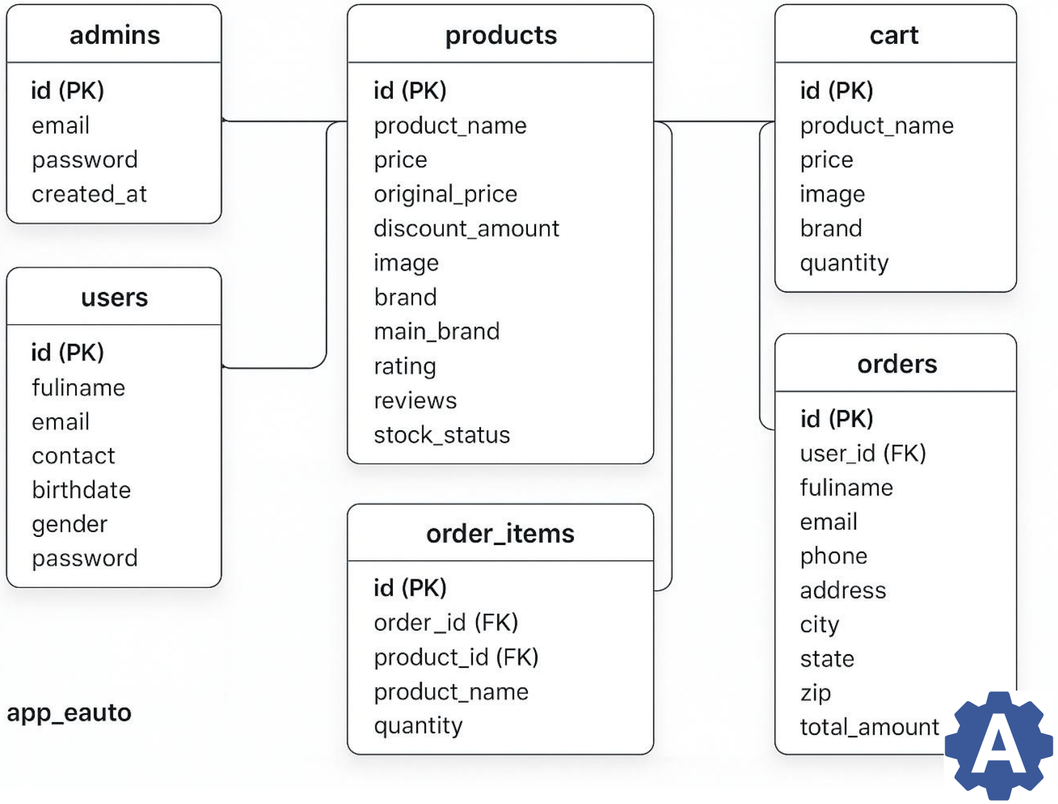
- **Technology Stack** – Frontend: HTML + CSS + Bootstrap + AOS, Backend: PHP, Database: MySQL.
- **Database Design** – ER Model, relationships, collections/tables structure.

ER Diagram Overview

- **Entities** – User, Product, Cart, Order, Payment, Delivery, Admin
- **Attributes** – e.g., User (UserID, Name, Email, Password), Product (ProductID, Name, Price, Stock)
- **Relationships** –
  1. User ↔ Order (One-to-Many)
  2. Order ↔ Product (Many-to-Many)
  3. Product ↔ Cart (Many-to-Many)
  4. Order ↔ Payment (One-to-One)
  5. Order ↔ Delivery (One-to-One)

Final ER Model

- Display the **full ER diagram** image (finalized version).
- Add a small caption: “This ER Model represents the logical database design for eAuto system, showing entities, attributes, and relationships.”



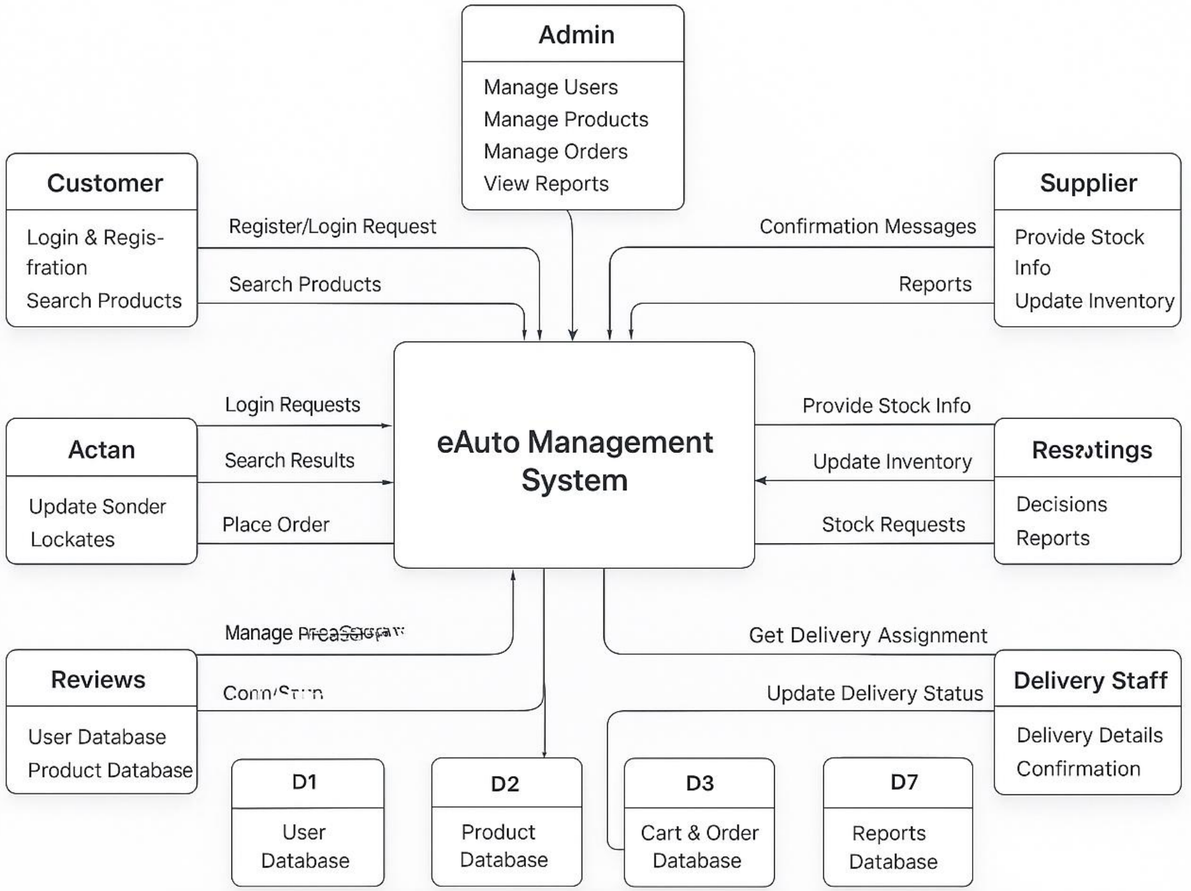
Context Level DFD (Overview)

- Show **single process box (System)** connected to:

1. User
2. Admin
3. Supplier
4. Payment Gateway

- Input/Output flows

Level 0 diagram image



Level 1 DFD (Explanation)

Processes (Inside System):

1.0 Login & Registration – Input: Login/Register request → Output: Confirmation → Data Store: D1 User DB

2.0 Product Management – Input: Product search/update → Output: Product info → Data Store: D2 Product DB

3.0 Cart & Order Management – Input: Add to Cart/Place Order → Output: Order confirmation → Data Store: D3 Cart & Order DB

4.0 Payment Processing – Input: Payment details → Output: Transaction confirmation → Data Store: D4 Payment Records

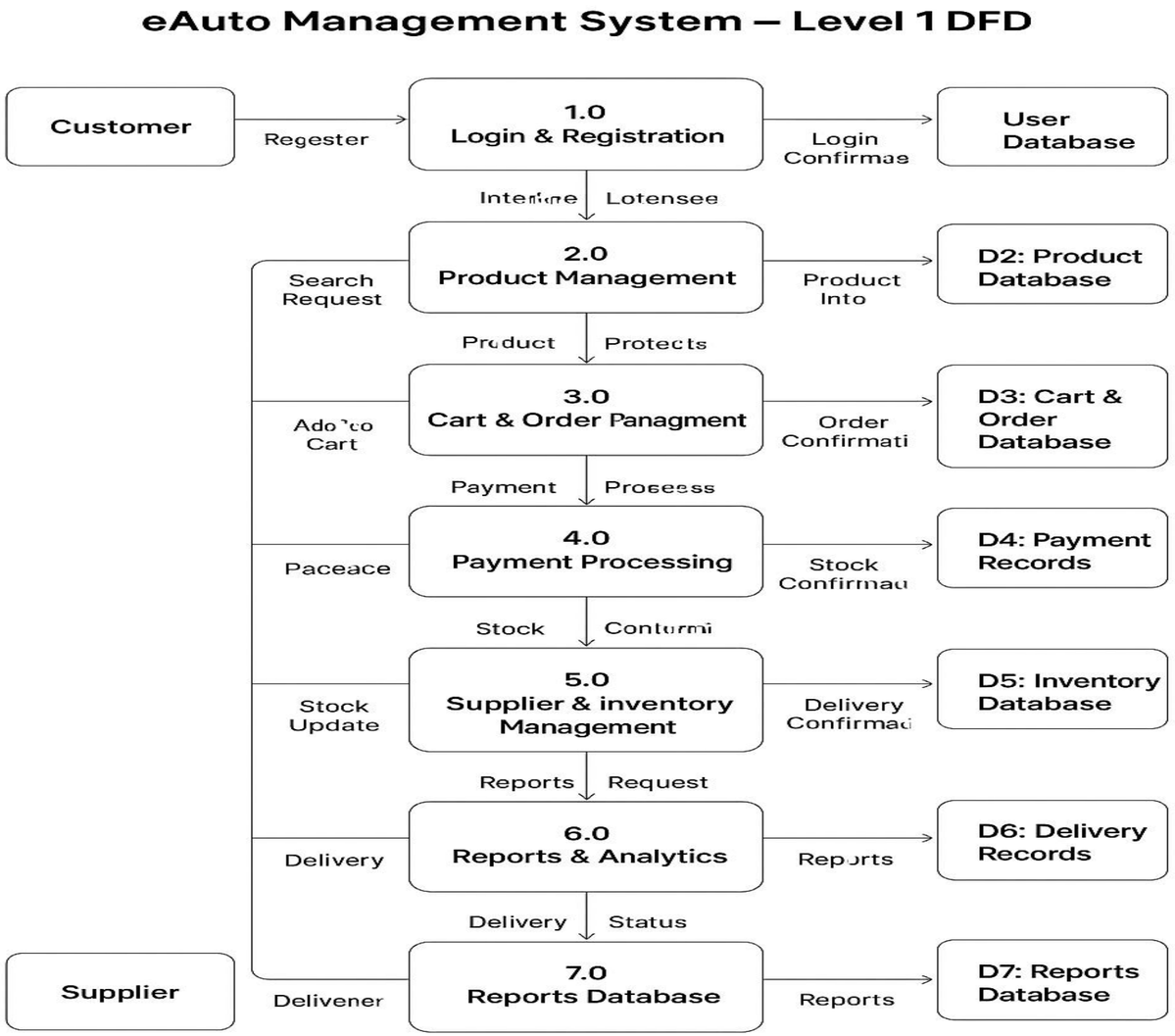
5.0 Supplier & Inventory Management – Input: Stock info update → Output: Stock confirmation → Data Store: D5 Inventory DB

6.0 Delivery Management – Input: Delivery assignment → Output: Delivery confirmation → Data Store: D6 Delivery Records

7.0 Reports & Analytics – Input: Report request → Output: Sales/Stock reports → Store: D7 Reports DB

Level 1 diagram image

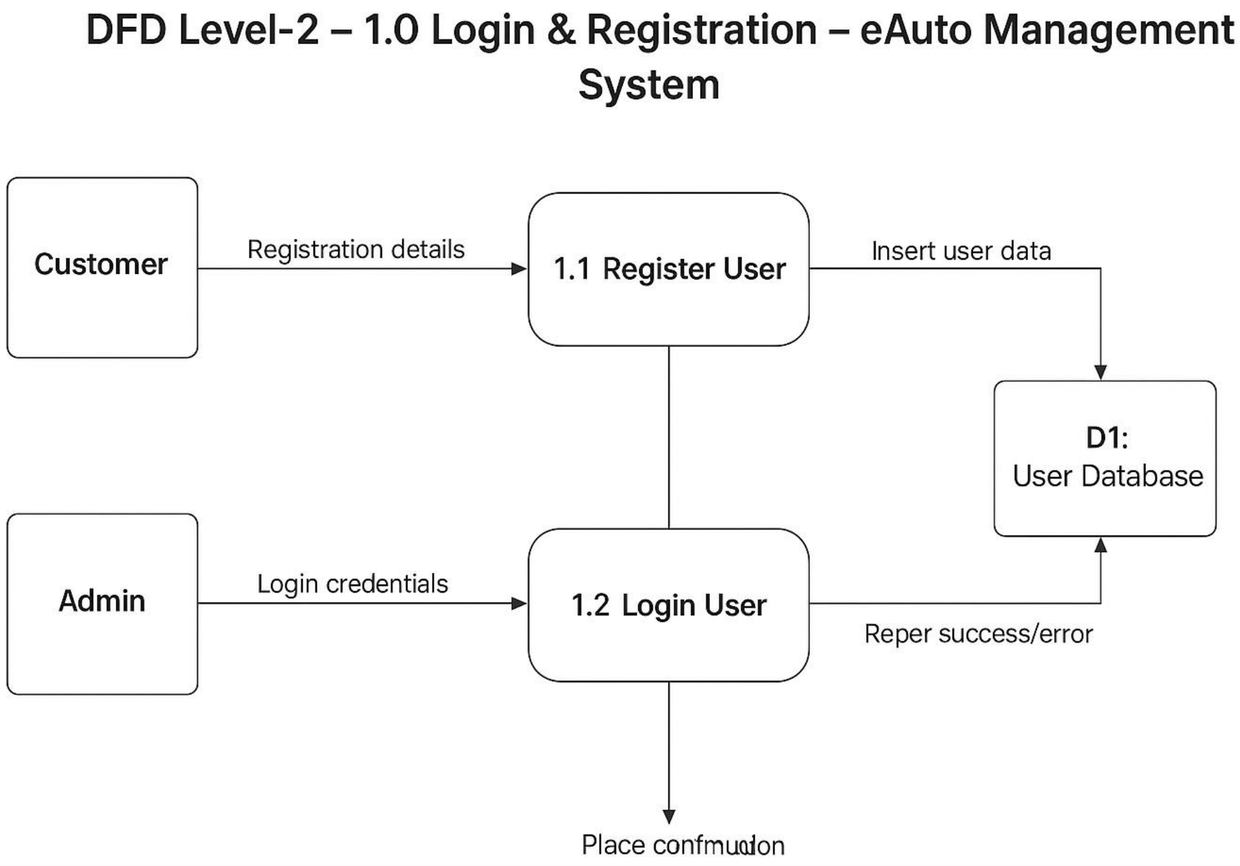
- External Entities (Customer, Admin, Supplier, Delivery Staff)
- Processes (1.0 – 7.0 inside system)
- Data Stores (D1–D7)



Process 1.0: Login & Registration

- User enters registration details / login credentials.
- System validates input against **User DB**.
- If valid → user session starts.
- If invalid → error message shown.
- Data stored in **User Collection (MongoDB)**.

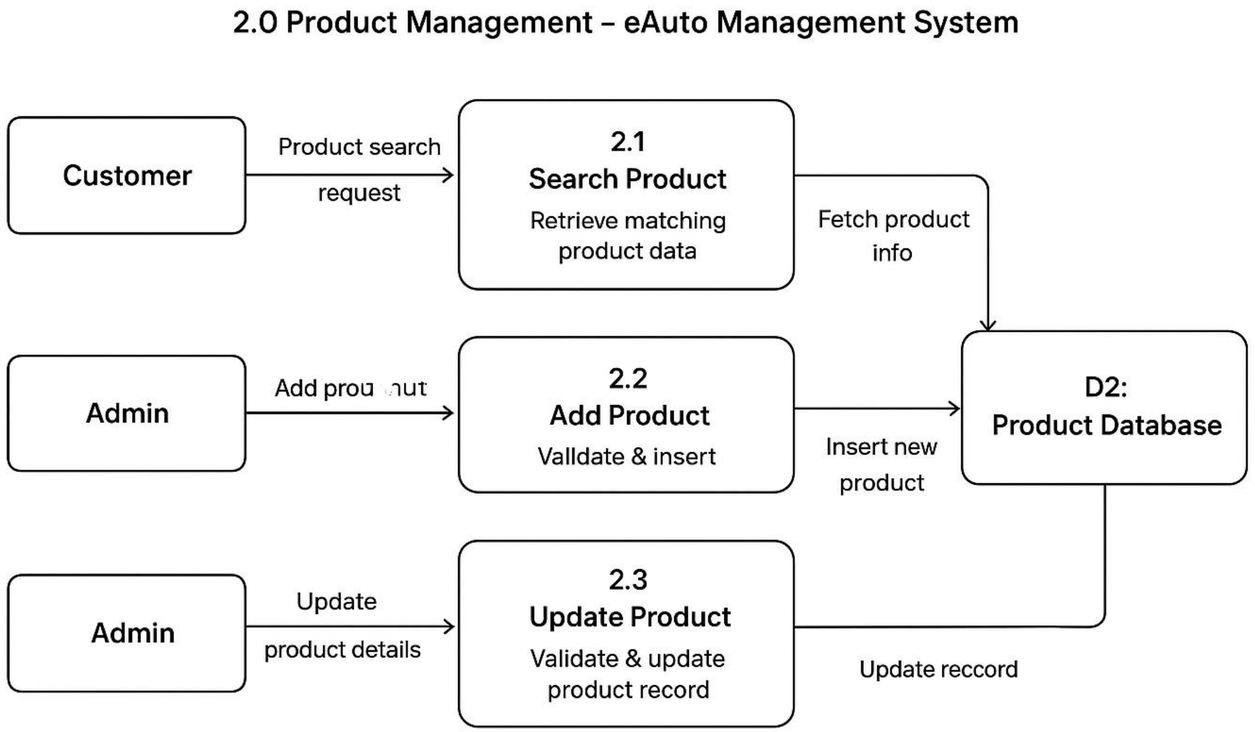
Level 1.0 diagram image



Process 2.0: Product Management

- User/Admin requests products.
- System queries **Product DB** (name, brand, category, price).
- Product details displayed on UI.
- Admin can **Add/Edit/Delete products**.
- Updates stored in **Product Collection**.

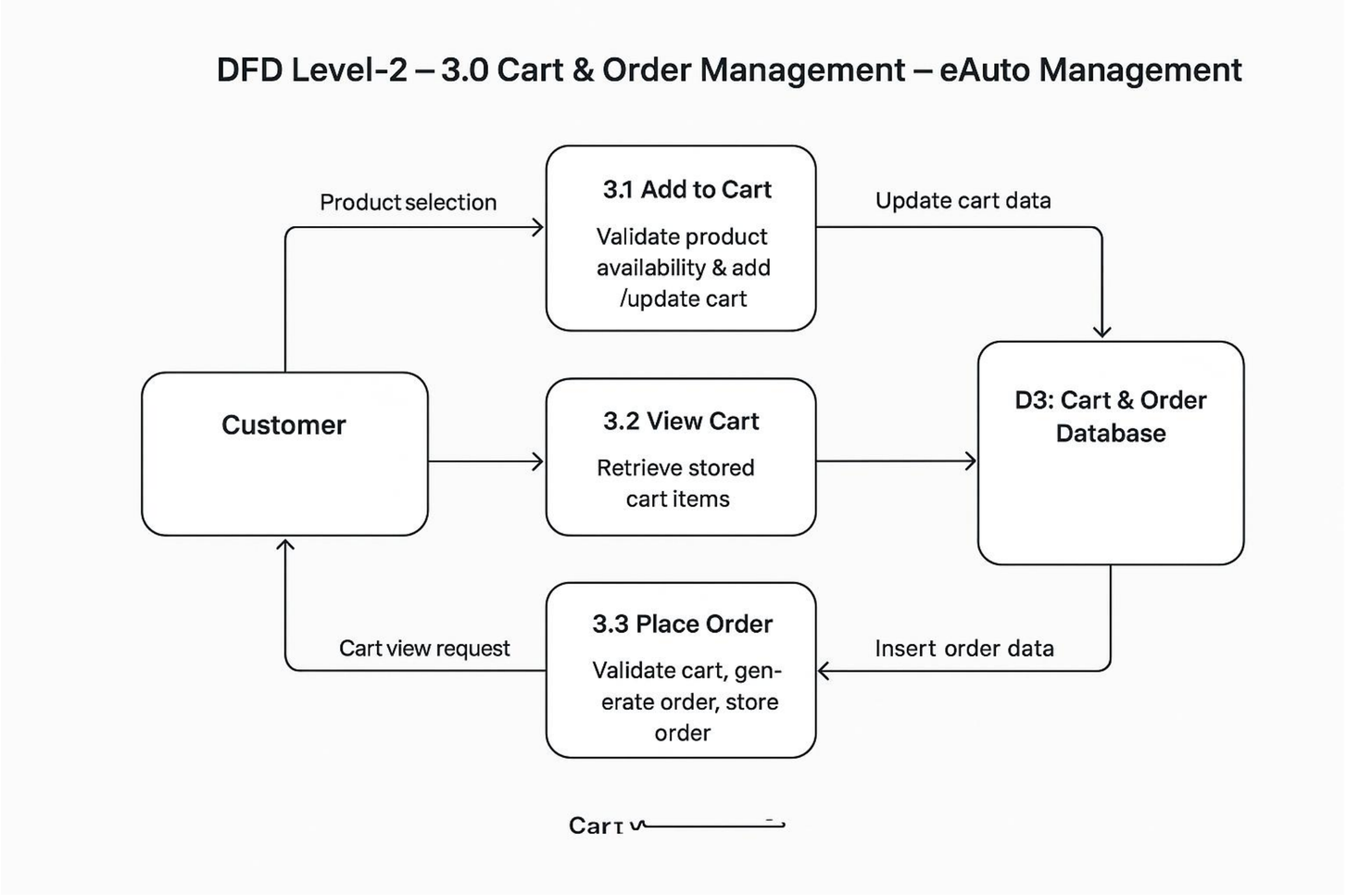
Level 2.0 diagram image



Process 3.0: Cart & Order Management

- User adds/removes products from cart.
- Cart data stored temporarily in **Cart DB**.
- On checkout → system generates order.
- Order stored in **Order DB** with user reference.
- Admin updates order status (Pending, Shipped, Delivered).

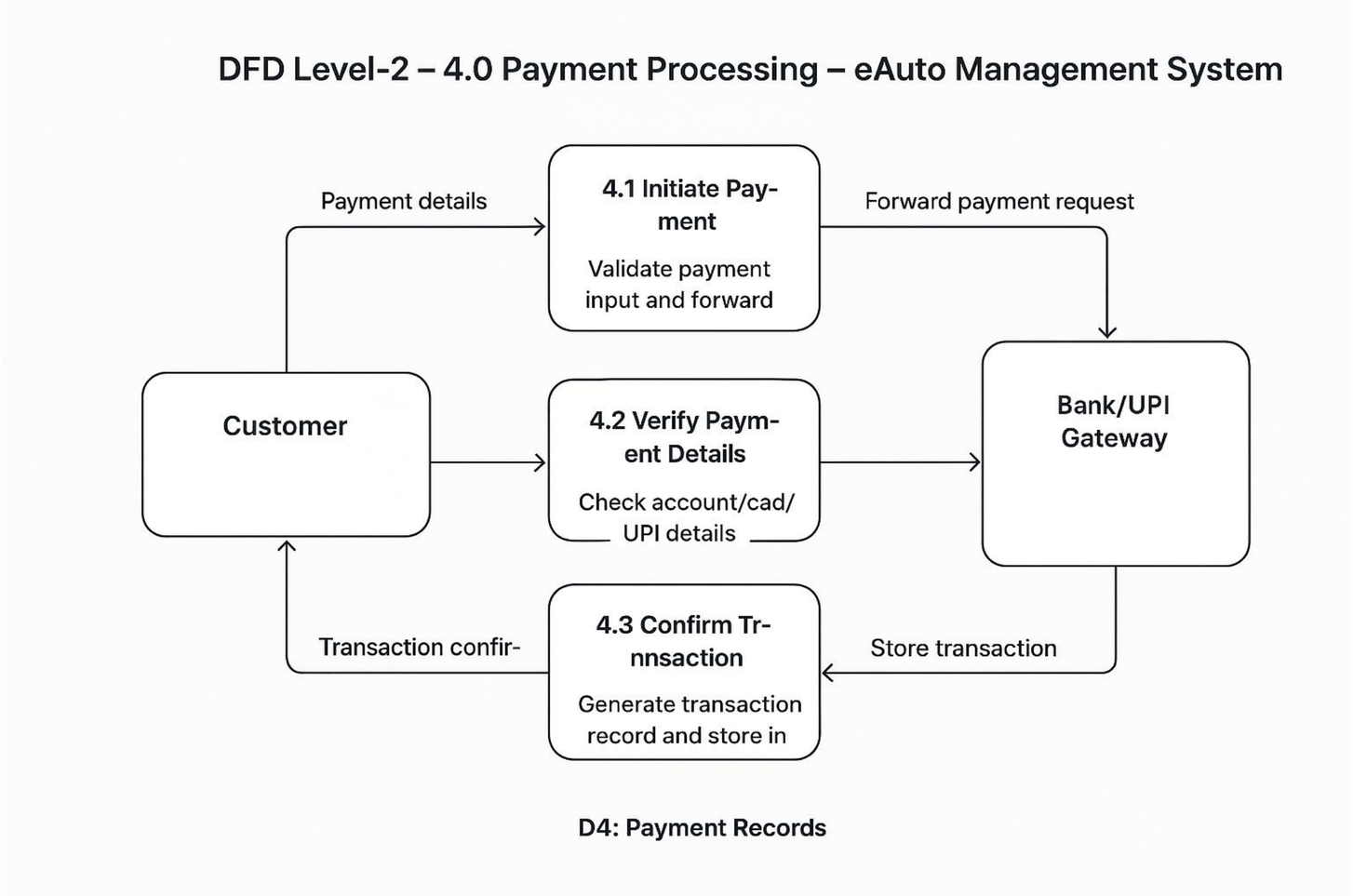
Level 3.0 diagram image



Process 4.0: Payment Processing

- User selects payment method (Cash, Card, UPI, NetBanking).
- System sends request to **Payment Gateway**.
- Gateway verifies details and returns status.
- If successful → confirmation stored in **Payment DB**.
- If failed → system requests re-try.

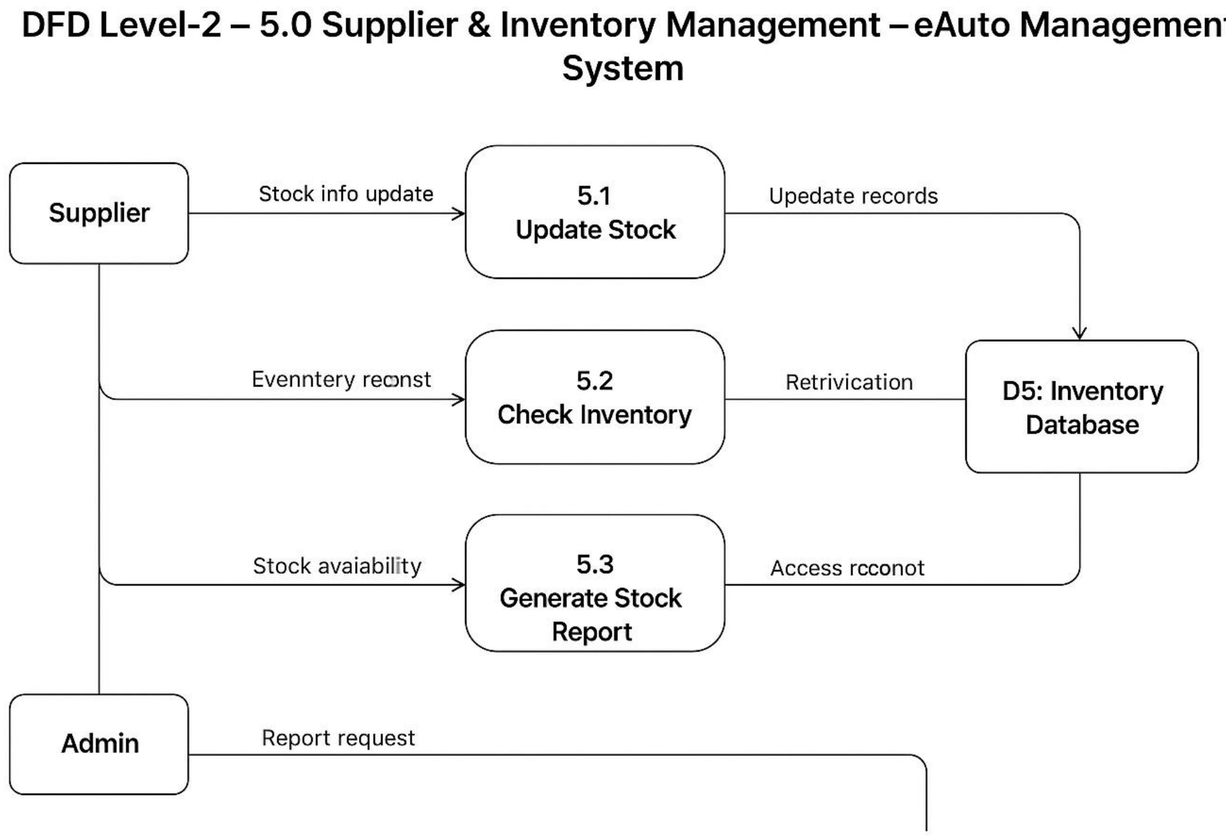
Level 4.0 diagram image



Process 5.0: Supplier & Inventory Management

- Admin checks product stock levels.
- Supplier updates stock into **Supplier DB**.
- System syncs product availability.
- Low stock alerts generated.
- Updates reflected in **Product DB**.

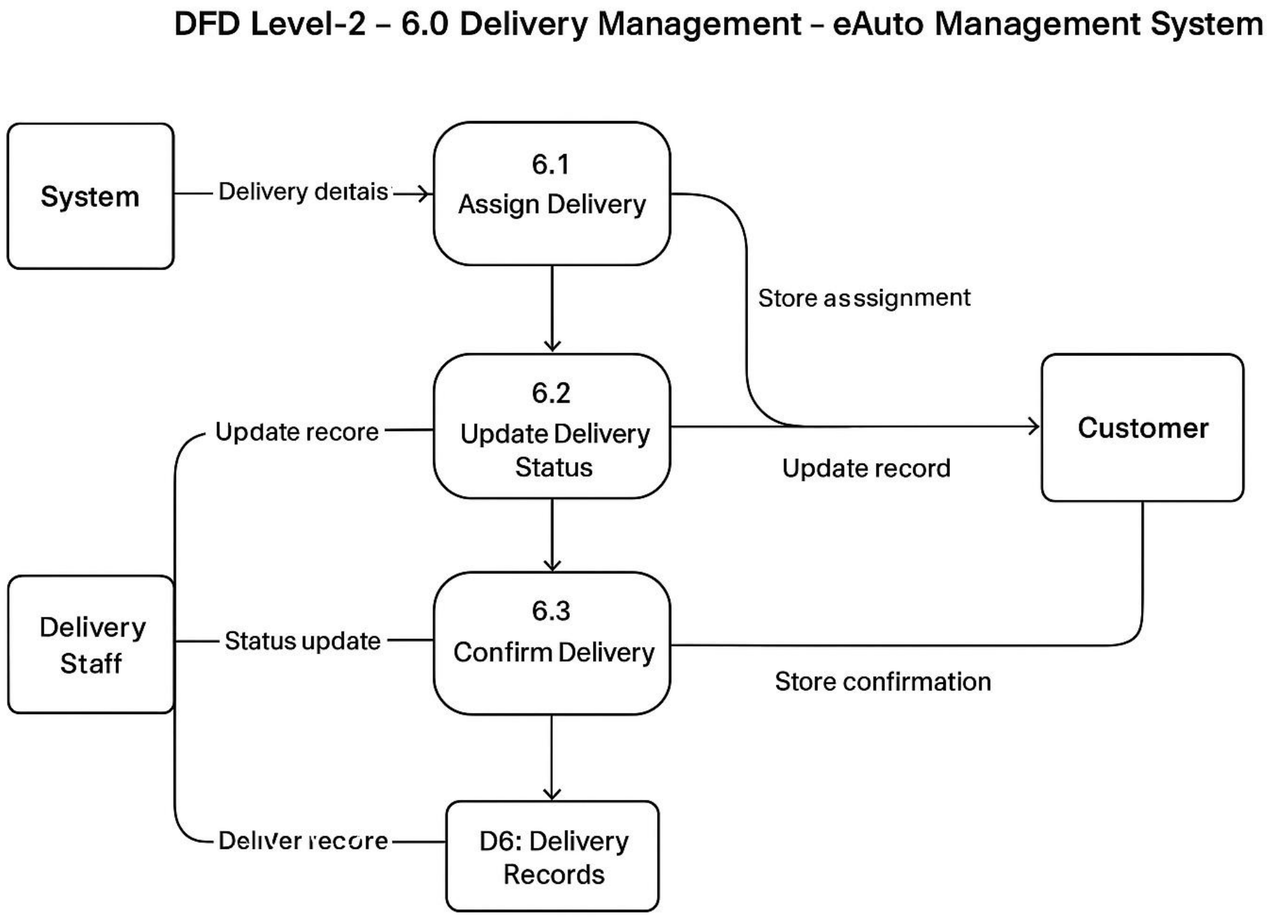
Level 5.0 diagram image



Process 6.0: Delivery Management

- Order assigned to delivery staff.
- Delivery details stored in **Delivery DB**.
- Staff updates status (Out for delivery, Delivered).
- User can track delivery in real time.
- System sends delivery confirmation.

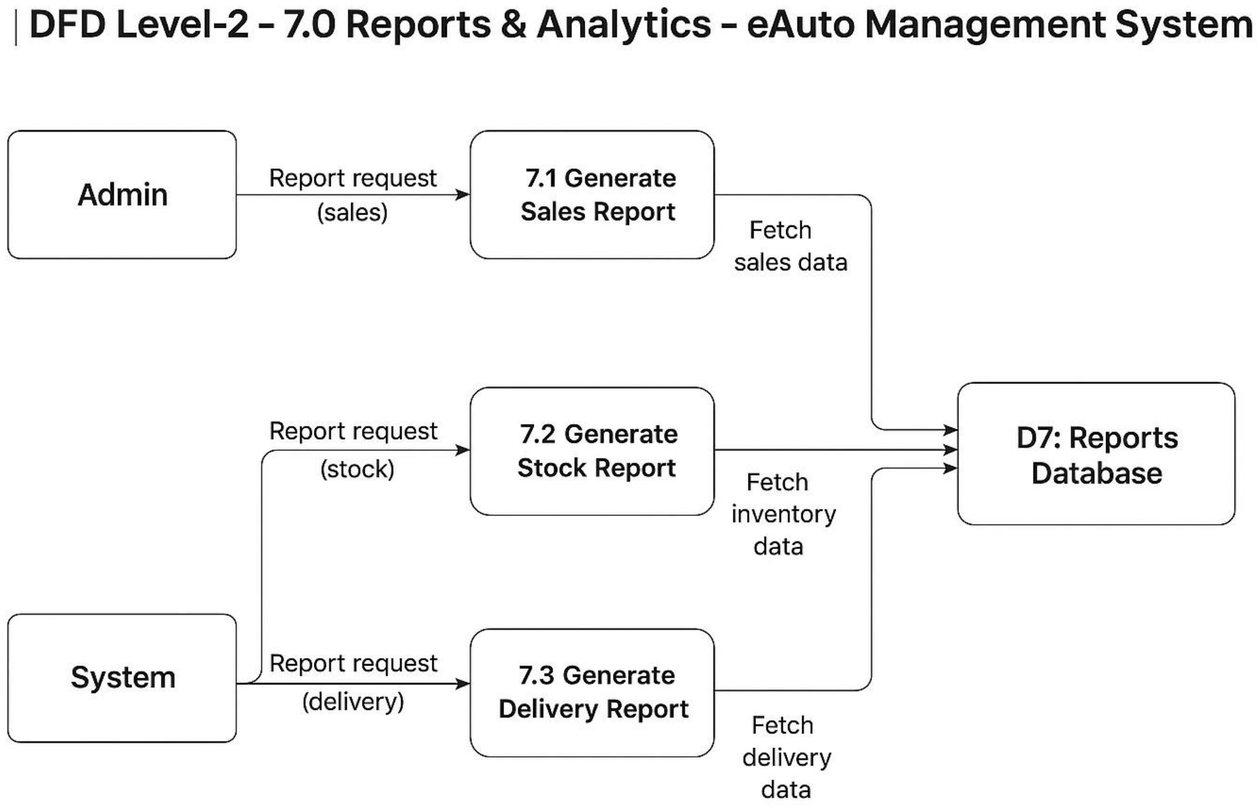
Level 6.0 diagram image



Process 7.0: Reports & Analytics

- Admin requests system reports.
- System collects data from **Orders, Users, Products DBs**.
- Generates **sales, stock & user reports**.
- Reports shown in **graphical / tabular form**.
- Data stored in **Reports Collection**.

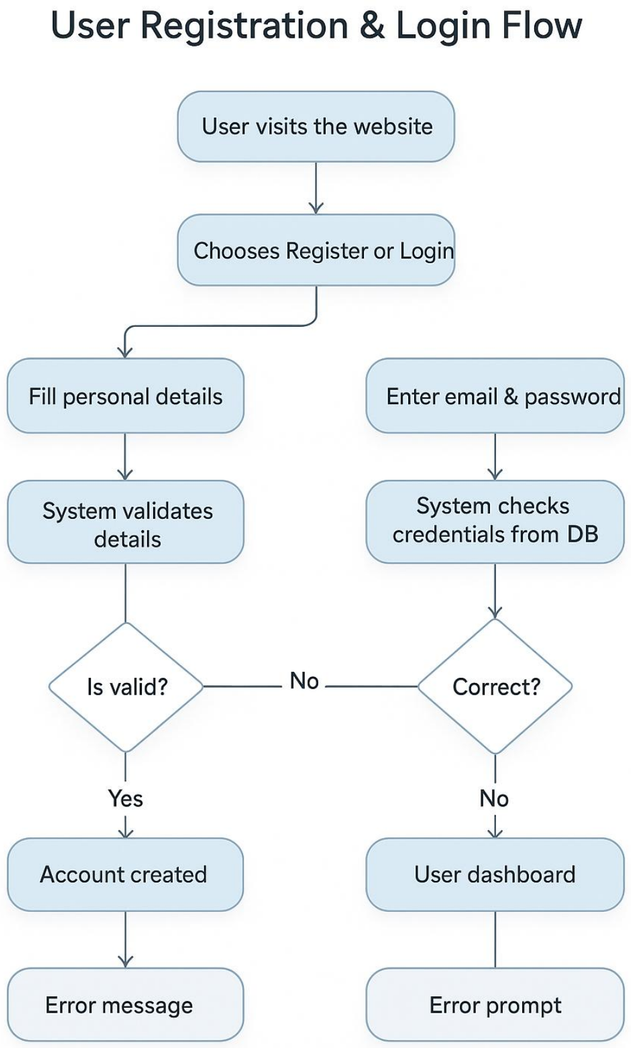
Level 7.0 diagram image



User Registration & Login Flow

- User enters registration/login details.
- System validates credentials against **User DB**.
- If valid → access granted, session starts.
- If invalid → error message shown.
- Data stored in **User Database**.

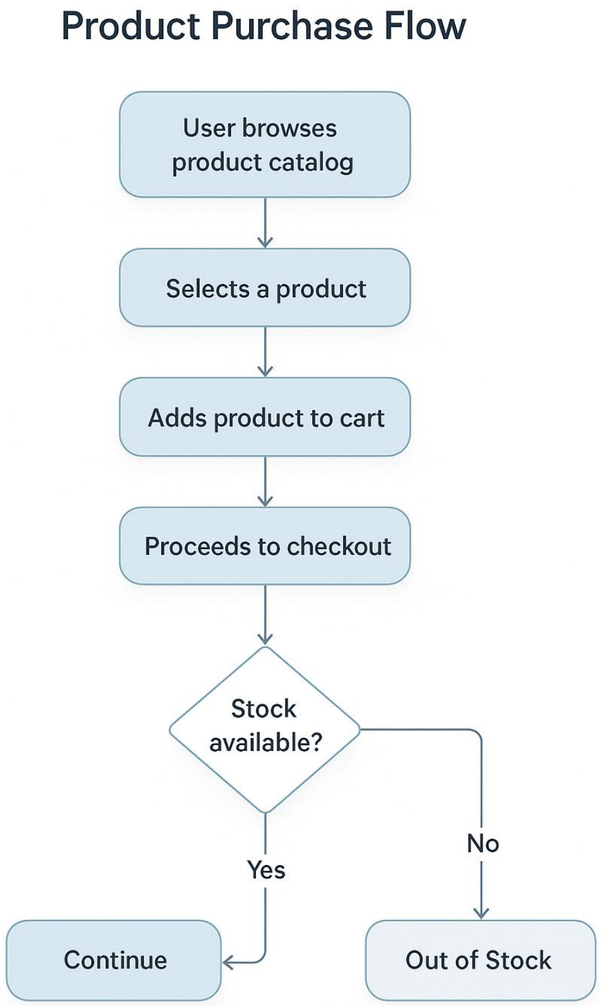
Flow Charts image



Product Purchase Flow

- User browses/searches products.
- Selects desired product and adds to cart.
- System checks stock availability.
- If available → proceed to checkout.
- If not available → show out-of-stock message.

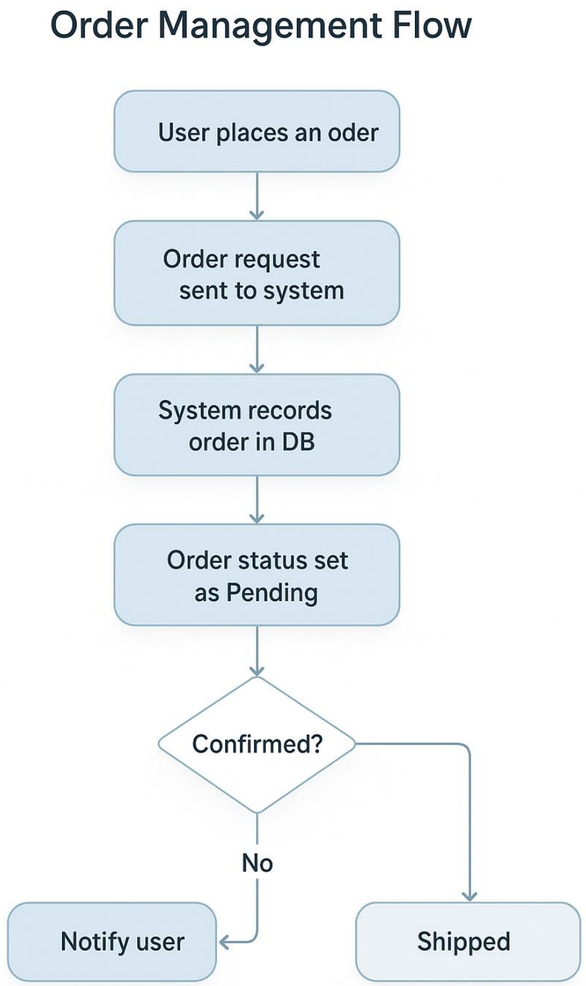
Flow Charts image



Order Management Flow

- User places order from cart.
- Order details stored in **Order DB**.
- Admin updates order status (Pending → Shipped → Delivered).
- User can track order progress.
- System notifies user at each stage.

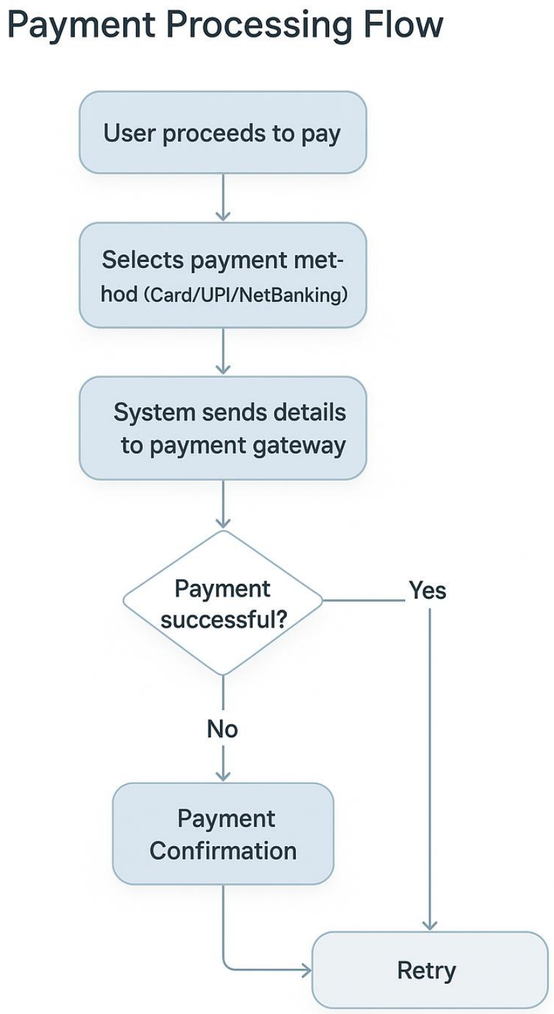
Flow Charts image



Payment Processing Flow

- User selects payment method (Cash, Card, UPI, NetBanking).
- System sends details to **Payment Gateway**.
- Gateway validates & returns success/failure.
- On success → confirmation stored in **Payment DB**.
- On failure → user prompted to retry.

Flow Charts image



Frontend Implementation

- **HTML** for structuring web pages
- **CSS** for styling and layout design
- **Bootstrap** for responsive and mobile-first design
- **AOS (Animate on Scroll)** for smooth animations.
- Provides interactive and mobile-friendly user experience.



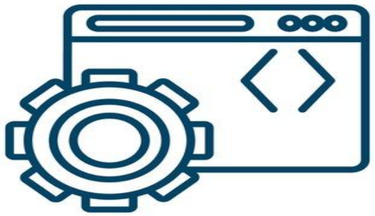




Backend & Database

Backend: Core PHP (mixed UI + business logic)

Database: **MySQL** (relational tables for Users, Products, Orders, etc.)



BACKEND



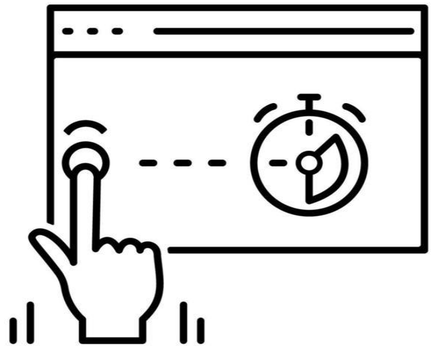
Security Features

- **Authentication & Session Management** for secure access.
- **Input Validation** to prevent errors & attacks.
- **Password Encryption** for user credentials.
- **Role-based Access** (User vs. Admin).
- Protects data integrity & confidentiality.

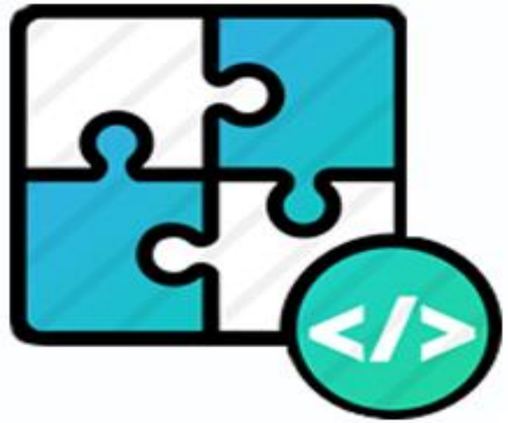


Test Cases (Unit & Integration)

- **Unit Testing:** Each module tested separately (Login, Cart, Checkout, etc.).
- **Integration Testing:** Check data flow between modules (Cart ↔ Order ↔ Payment).
- Ensures modules work correctly both individually and together.
- Detects errors early in development.



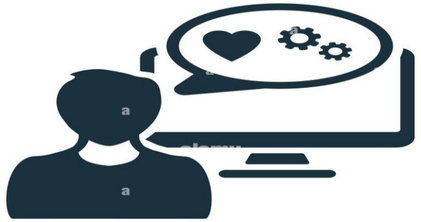
UNIT TESTING



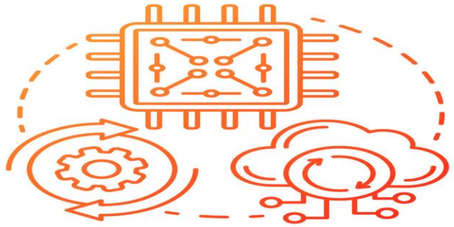
Integration Testin

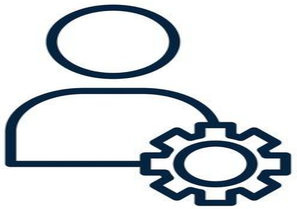
User & Admin Testing

- **User Testing (UI/UX):** Verify navigation, product browsing, cart, and order tracking.
- **Functional Testing:** Validate real-time actions like add-to-cart, checkout, payment.
- **Admin Testing:** Product management, order updates, report generation.
- Confirms system works as expected for both roles.



USER TESTING

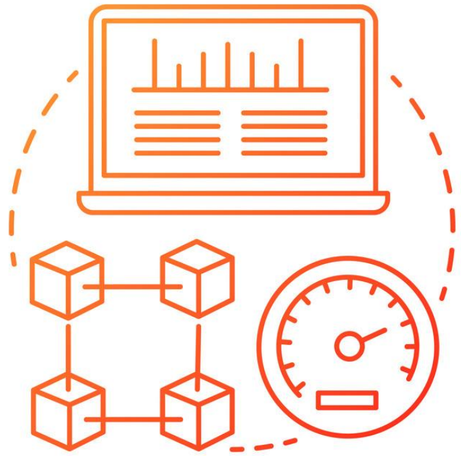




ADMIN
STRATEGY

Performance & Load Testing

- Test system speed under normal and peak loads.
- Verify response time for product search, cart, and payment.
- Load testing with multiple users simultaneously.
- Ensures scalability and reliability of the system.
- Detects bottlenecks before deployment.



Limitations of the Project

◆ **Technical Limitations**

- Tight coupling of PHP with HTML/CSS made the codebase less modular and harder to maintain
- MySQL database lacked flexibility for handling unstructured data

◆ **Functional Limitations**

- Limited payment options (Cash, UPI, Card only).
- Reports are simple, not AI-driven.

Future Enhancements

- Mobile App version (Android & iOS).
- AI-based product recommendations.
- Advanced reporting with data visualization dashboards.
- Integration with third-party delivery partners (Swiggy Genie, Dunzo, etc.).
- Multi-language and multi-currency support for wider audience.

Conclusion (Summary & Achievements)

◆ **Summary of Work**

- Designed & implemented eAuto – Bike Spare Parts E-Commerce Platform.
- Frontend with HTML/CSS, Bootstrap & AOS for UI/UX.
- Backend with PHP APIs integrated with MySQL.
- Features include login, product browsing, cart, orders, payment, and delivery tracking.

◆ **Achievements**

- Fully functional e-commerce workflow (User ↔ Admin ↔ Supplier ↔ Delivery).
- Secure login & session management.
- Simplified cart and order management system.
- Automated stock & delivery status updates.

Conclusion (Future Scope)

- Develop dedicated mobile app (Android/iOS).
- AI-based recommendations for personalized shopping.
- Advanced analytics dashboards for Admin.
- Multi-vendor & franchise support.
- Integration with external payment & delivery APIs.

References











